

Problem 2

Problem 2

Problem 1 Let $f(x)$ and $g(x)$ be functions for which we only know the following:

$$\int_1^4 f(x) dx = 7 \quad \int_2^4 f(x) dx = 5 \quad \int_1^4 g(x) dx = 2$$

Compute the following integrals, if possible. If it is not possible, give examples explaining why not.

(a) $\int_1^4 (8f(x) - 7g(x)) dx$

Explanation.

$$\begin{aligned} \int_1^4 (8f(x) - 7g(x)) dx &= 8 \int_1^4 f(x) dx - 7 \int_1^4 g(x) dx \\ &= 8(7) - 7(2) \\ &= 56 - 14 = 42 \end{aligned}$$

(b) $\int_1^2 (-f(x)) dx$

Explanation. First notice that

$$\int_1^4 f(x) dx = \int_1^2 f(x) dx + \int_2^4 f(x) dx$$

Therefore,

$$\int_1^4 f(x) dx - \int_2^4 f(x) dx = \int_1^2 f(x) dx$$

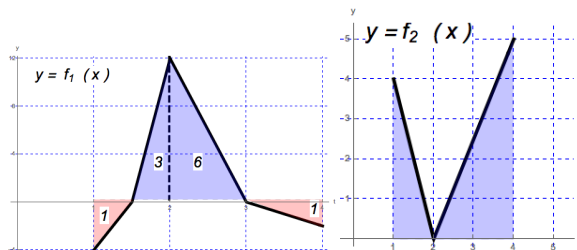
So

$$\begin{aligned} \int_1^2 (-f(x)) dx &= - \int_1^2 f(x) dx \\ &= - \left(\int_1^4 f(x) dx - \int_2^4 f(x) dx \right) \\ &= -(7 - 5) = -2. \end{aligned}$$

(c) $\int_1^4 |f(x)| dx$

Problem 2

Explanation. We are not given enough information to solve this integral because we do not know the regions where f is positive or negative. Consider the following two functions $f_1(x)$ and $f_2(x)$:



Just using geometry, one can check that

$$\int_1^4 f_1(x) dx = 7 \quad \int_2^4 f_1(x) dx = 5 \quad \int_1^4 f_2(x) dx = 7 \quad \int_2^4 f_2(x) dx = 5$$

and so both f_1 and f_2 satisfy the assumptions of f . But notice that

$$\int_1^4 |f_1(x)| dx = 11 \quad \text{and} \quad \int_1^4 |f_2(x)| dx = 7$$

These two examples demonstrate that we were not given enough information to solve this problem.

(d) $\int_1^4 (2 - x + f(x)) dx$

Explanation. First notice that since the integral is linear over addition:

$$\int_1^4 (2 - x + f(x)) dx = \int_1^4 2 dx - \int_1^4 x dx + \int_1^4 f(x) dx = \int_1^4 2 dx - \int_1^4 x dx + 7. \quad (1)$$

By using geometry, we can see that

$$\int_1^4 2 dx = 2(4 - 1) = 6$$

$$\int_1^4 x dx = 1(4 - 1) + \frac{1}{2}(4 - 1)(4 - 1) = 3 + \frac{9}{2} = \frac{15}{2}.$$

Then substituting into equation (1) gives:

$$\int_1^4 (2 - x + f(x)) dx = 6 - \frac{15}{2} + 7 = \frac{11}{2}.$$